

Assignment 3: Countdown Timer Application

Introduction

In this assignment, you will create a simple countdown application. The task involves using CSS to design an interface with an apple in the center of the screen, containing an apple and three buttons. These buttons control the start, stop, and pause functionalities of the timer. The timer runs for 1 minute. The timer will change its CSS style after it reaches 15 seconds. After the countdown finished, the clock will alert the user to take a short break.

Objective

- Develop a basic HTML page featuring a paragraph for displaying the timer, two buttons for controlling the timer, and an apple image as the background.
- Utilise CSS to style the page with a light blue background color, green start button, red stop button, and orange pause button. The timer text should be white by default and turn yellow on hover. When the timer reaches 15 seconds, the text color must change to red.

Task Description

The tasks outlined below provide detailed steps and requirements to guide you through completing the assignment.

Task 1: HTML (Hypertext Markup Language) (30%)

Create a basic HTML file with the layout of a countdown timer.

Requirements:

- Boilerplate (10%): Implement standard HTML boilerplate code to structure each page, including DOCTYPE, HTML, HEAD, BODY, etc. Make sure the navigation bar at the top of the webpage includes a "Timer" Button.
- Buttons and Timer (10%): Create a paragraph with the id "timer" to display the timer and add buttons for starting, stopping, and pausing the timer.

- Apple Image (10%): Add a background image of apple, you can make it by yourself or download from the website. The apple should be horizontally centred in the webpage.
- You can use this link to remove background for the images:
<https://www.remove.bg/>

Task 2: CSS (Cascading Style Sheets) (30%)

Style the buttons, timer, and page appropriately for the application using CSS.

Requirements:

- Boilerplate (5%): Create a styles.css file and link it to the HTML file. Include a style to make the background of the page a light blue and center the countdown timer on the page.
- Buttons (10%): Style each button with the Roboto font family, white font color. Background color should be green for start, red for stop, and orange for pause. Apply a black border when hovering over each button.
- Apple (15%): The timer text in the apple should be Orbitron font. When hovered over, the timer text should become yellow; otherwise, the timer text should become white. Additionally, when the timer goes below 15 second, change the text color to red and add a background color with rounded corners of 5px to make it stand out from the apple.

Task 3: JS (JavaScript) (40%)

Implement timer functionality using JavaScript.

Requirements:

- Timer (15%): Develop a timer functionality that tracks the time since the start button was pressed. When the timer reaches 0, display an alert to take a break. Pressing the stop button resets the timer, which only starts again when the start button is clicked.
- Pause Button (15%): Implement functionality to pause the timer instead of resetting it. When the pause button is clicked, the text should switch to "Resume", and clicking it again should return to "Pause" while the timer continues running.

- Disable button (10%): Initially, the "Pause" and "Stop" buttons should be disabled and the "start" button should be enabled. After the "Start" button is clicked, the "Pause" and "Stop" buttons should be enabled and the Start" button should be disabled. After "Stop" button is clicked, the three buttons should be reverted back to the initial state (the "Pause" and "Stop" buttons should be disabled and the "Start" button should be enabled).

Hint: How to import font in CSS?

<https://support.secureprivacy.ai/articles/238540-how-to-add-custom-fonts-using-css-in-secure-privacy-banner>

- Use the @import rule: Add the following line at the very beginning of your custom CSS code to import your chosen font:

```
@import url('https://fonts.googleapis.com/css2?family=Your+Font+Name&display=swap');
```

- Replace Your+Font+Name with the actual font name from the font library. For example, Roboto would look like

```
@import url('https://fonts.googleapis.com/css2?family=Roboto&display=swap');
```

Important: Always place font imports at the very start of your CSS to ensure they load correctly.

Deliverables

Submit a .ZIP file to Moodle containing the following items:

- index.html
- styles.css
- script.js
- apple image
- screenshot of application